

ULYANOVSK, Russian Federation

WMO: 277850

Lat: 54.276N

Lon: 48.243E

Elev: 141

StdP: 99.64

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.9	-25.5	-31.8	0.2	-28.7	-28.2	0.3	-25.3	12.1	-1.9	11.0	-4.4	2.2	340	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.8	19.9	29.8	19.7	27.9	18.8	21.6	28.0	20.7	27.3	19.8	26.1	4.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.4	24.6	18.5	13.6	23.6	17.6	12.8	22.9	63.5	28.0	60.4	27.3	57.4	26.1	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	10.0	9.0	-32.0	33.9	4.0	2.1	-34.9	35.4	-37.3	36.6	-39.6	37.8	-42.5	39.4
			-32.0	23.1	4.0	1.2	-34.9	24.0	-37.2	24.7	-39.5	25.3	-42.3	26.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-10.6	-10.6	-4.1	6.1	13.5	18.0	20.3	18.3	12.6	5.5	-1.8	-7.9	(6)
(7)		DBStd	12.41	7.41	7.57	5.49	5.43	4.88	4.08	3.50	4.02	4.43	4.48	5.48	7.03	(7)
(8)		HDD10.0	2901	639	576	436	142	22	1	0	0	22	152	355	556	(8)
(9)		HDD18.3	5053	898	809	694	368	161	54	20	52	179	398	605	814	(9)
(10)		CDD10.0	1087	0	0	0	24	132	242	320	258	99	12	0	0	(10)
(11)		CDD18.3	197	0	0	0	0	12	45	82	51	7	0	0	0	(11)
(12)		CDH23.3	2076	0	0	0	7	172	473	825	534	65	0	0	0	(12)
(13)		CDH26.7	674	0	0	0	0	41	144	279	199	12	0	0	0	(13)
(14)	Wind	WSAvg	4.0	4.1	4.1	4.2	4.3	4.2	3.8	3.3	3.5	3.8	4.3	4.5	4.2	(14)
(15)	Precipitation	PrecAvg	496	34	27	25	32	45	68	50	51	55	41	35	33	(15)
(16)		PrecMax	695	78	53	61	95	108	124	147	103	152	75	63	80	(16)
(17)		PrecMin	338	13	10	4	2	2	1	6	15	10	17	10	11	(17)
(18)		PrecStd	90	16	12	13	21	28	36	34	26	38	18	14	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	2.9	10.4	24.2	30.1	33.4	34.2	34.9	28.3	19.8	11.1	4.2	(19)
(20)		MCWB	1.9	1.5	6.1	13.4	17.8	19.6	19.5	19.9	16.9	12.1	9.2	3.4	(20)	
(21)		2%	DB	1.4	1.9	6.0	21.2	27.6	30.3	31.6	31.1	25.1	16.7	8.5	2.6	(21)
(22)		MCWB	0.7	0.9	3.3	11.9	16.5	19.6	20.2	19.7	15.9	11.2	7.1	1.7	(22)	
(23)		5%	DB	0.5	1.0	3.9	18.3	25.1	28.0	29.8	28.8	22.8	14.2	6.4	1.6	(23)
(24)		MCWB	-0.2	0.2	1.8	10.8	15.5	18.9	20.3	19.3	15.2	10.6	5.3	0.9	(24)	
(25)		10%	DB	-0.8	-0.2	2.5	15.4	22.7	26.0	27.8	26.1	20.0	12.0	4.8	0.7	(25)
(26)		MCWB	-1.5	-1.1	1.0	9.3	14.2	18.2	19.5	18.4	14.3	9.1	3.9	0.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	2.0	6.6	14.6	19.7	22.0	23.1	22.2	18.7	13.7	9.7	3.8	(27)
(28)		MCDWB	2.3	2.5	10.3	22.4	28.5	28.6	29.4	28.7	25.2	16.9	10.9	4.0	(28)	
(29)		2%	WB	0.7	1.0	3.4	12.8	17.6	20.9	22.0	20.9	17.1	12.0	7.4	1.8	(29)
(30)		MCDWB	1.2	1.7	5.5	19.4	25.0	27.6	27.9	28.1	22.7	15.4	8.4	2.3	(30)	
(31)		5%	WB	-0.1	0.2	2.1	11.2	16.3	19.9	21.1	19.9	15.9	10.6	5.4	0.9	(31)
(32)		MCDWB	0.5	1.0	3.5	17.3	23.3	26.1	27.4	26.9	20.8	13.6	6.2	1.4	(32)	
(33)		10%	WB	-1.5	-1.1	1.2	9.6	14.9	18.9	20.2	18.9	14.8	9.5	3.8	0.1	(33)
(34)		MCDWB	-0.8	-0.2	2.3	15.1	21.2	24.6	26.2	25.4	19.4	11.9	4.6	0.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	6.5	7.7	8.2	10.0	11.5	11.0	11.0	11.0	9.6	7.3	4.9	5.7	(35)	
(36)		5% DB	MCDBR	4.7	5.2	8.2	14.4	15.2	14.2	14.1	15.1	13.5	11.2	6.2	4.1	(36)
(37)		MCWBR	4.7	4.7	5.8	7.6	7.0	6.3	5.7	6.5	6.6	7.0	5.2	4.0	(37)	
(38)		5% WB	MCDBR	4.7	5.0	7.1	13.3	13.4	12.1	12.0	13.0	11.9	9.9	5.5	4.0	(38)
(39)		MCWBR	4.7	4.7	5.2	7.4	6.7	6.0	5.6	6.4	6.4	6.8	5.0	3.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.319	0.359	0.417	0.393	0.385	0.400	0.425	0.375	0.350	0.316	0.287	(40)	
(41)		taud	2.092	2.060	2.044	2.124	2.320	2.382	2.339	2.253	2.372	2.346	2.236	2.087	(41)	
(42)		Ebn at Noon	631	740	796	803	853	864	840	790	789	717	603	539	(42)	
(43)		Edh at Noon	68	100	128	136	118	113	116	119	93	76	60	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.80	1.79	3.20	4.23	5.55	5.94	5.89	4.71	3.00	1.47	0.62	0.45	(44)	
(45)		RadStd	0.12	0.22	0.37	0.39	0.41	0.56	0.51	0.42	0.41	0.25	0.10	0.12	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	+1.28	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	N/A	(47)

Nomenclature: See separate page